PSYC 4503 Group Project Details

The goal of this project is for you to apply the core concepts of psychometrics by developing a brief psychological scale from scratch. You will move through the same steps used by researchers to create reliable and valid instruments: 1) Defining a construct, 2) writing test items, 3) testing them, 4) analyzing your data, and 5) refining your scale. At the end of the semester, each group will present their findings to the class.

Project Goals

By the end of the project, you will:

- Understand the steps in creating a psychological test.
- Demonstrate reliability and preliminary validity using real data.
- Evaluate and revise your scale based on psychometric principles.
- Communicate findings in both written and oral formats.

Overview of Project Components

This project is made up of several mini-assignments, which are due throughout the semester. As a group, you will complete the following:

1. Project Proposal (Due 9/11)

The project proposal is a 2-page write-up of your project which will cover the following:

1) You will select one broad, psychological construct (e.g., depression – since I'm using it as an example, you can't pick depression as your topic). You will need to provide a formal, operational definition of this construct.
2) For this construct, you will then select **three** factors that you think make up your construct (e.g., lack of energy, sadness, hopelessness)
3) Provide a summary of 3 scholarly sources. These sources should provide evidence for why your three factors contribute to your larger psychological construct.

2. Preliminary Item Pool (Due 9/25)

- You will need to create 30, original test items that you will use to measure your construct. You should have 10 items that correspond to each factor (think of it as making 3, ten-item scales (so to use the earlier example, 10 items measuring lack of energy, 10 measuring sadness, and 10 measuring hopelessness. These three subscales combine to make a larger 30-item depression scale).
- You should include the items AND the specific scale you plan to use (e.g., 5-point Likert; see the example on D2L). Use the same scale for all items! (e.g., don't mix Likert scale and short answer questions).

3. Pilot Testing (Weeks 8-10, data collection closes on 10/30)

This is where the fun begins. You'll need to administer your scale to **at least 30 participants** (try to get as many respondents as possible!). This can be friends, family, classmates, etc.

- Collect basic demographics (Probably 3-5 questions – I'll let you decide what demographic info you think is important to measure)
- Collect responses to your 30 scale items

4. Data Analysis

This part is also fun! You'll need to fire up SPSS and analyze your data. The computers in OD 126 all have SPSS installed and are available for you to use on your own time. You will need to report the following:

- Descriptive statistics for each subscale (mean, sd, min, max)
- Descriptive statistics for your demographics
- Cronbach's alpha (internal consistency) (overall for the full scale)
- Cronbach's alpha for each subscale (if you're counting, that's four separate alphas!)
- Correlations between subscales (convergent validity)

Unfortunately, we won't have time to cover SPSS in class. However, you should have gotten an introduction to it in your stats class. I'll also post some guides on D2L as things get closer (these will be posted by Week 10).

5. Group Presentation (Week 15)

As a group, you will present a 15-minute PowerPoint talk summarizing your project. This presentation should include:

- An overview of your scale, including the background literature (you should include at least four peer reviewed sources)
- An overview of your methods (participants, materials (including examples of your scale items), and procedure)
- A summary of your results (descriptives, reliability, and validity)
- A general discussion (summary of main findings plus key takeaways)
- A conclusion